



الهيئة الاتحادية للهوية  
والجنسية والجمارك وأمن المنافذ  
FEDERAL AUTHORITY FOR IDENTITY,  
CITIZENSHIP, CUSTOMS & PORT SECURITY

# ID Card Toolkit iOS and Mac Sample Build Instructions

Version 1.4

March 2026

CONFIDENTIAL

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# Table of Contents

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|                                                           |   |
|-----------------------------------------------------------|---|
| Revision History .....                                    | 3 |
| Abbreviations .....                                       | 4 |
| 1 Introduction .....                                      | 5 |
| 1.1 Purpose .....                                         | 5 |
| 2 iOS and Mac Sample Application Build Instructions ..... | 6 |
| 2.1 Prerequisites .....                                   | 6 |
| 2.2 Build Instructions .....                              | 6 |
| 2.2.1 Step 1: Obtain Open-Source Libraries .....          | 6 |
| 2.2.2 Step 2: Open the Sample Project .....               | 6 |
| 2.2.3 Step 3: Configure Search Paths .....                | 6 |
| 2.2.4 Step 4: Build .....                                 | 6 |
| 2.2.5 Step 5: Connect a Device .....                      | 6 |
| 2.2.6 Step 6: Code Signing .....                          | 7 |
| 2.2.7 Step 7: Run .....                                   | 7 |
| 2.2.8 Step 8: Configure the Toolkit .....                 | 7 |

## Revision History

| Version | Date           | Description of Changes                                                       |
|---------|----------------|------------------------------------------------------------------------------|
| 1.0     | 17th May, 2018 | Release Version 1.0.5                                                        |
| 1.1     | 05th Jun, 2018 | Release Version 1.0.6                                                        |
| 1.2     | 15th Feb, 2019 | Release Version 1.0.13                                                       |
| 1.3     | 20th Dec, 2023 | Release Version 3.0.0                                                        |
| 1.4     | Mar 2026       | Toolkit v3.0.5. Updated Xcode instructions. Improved configuration guidance. |

## Abbreviations

| Abbreviation | Description                                                          |
|--------------|----------------------------------------------------------------------|
| ICP          | Federal Authority for Identity, Citizenship, Customs & Port Security |
| IDE          | Integrated Development Environment                                   |
| iOS          | iPhone Operating System                                              |
| SDK          | Software Development Kit                                             |
| SP           | Service Provider                                                     |
| URL          | Uniform Resource Locator                                             |

# 1. Introduction

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The Federal Authority for Identity, Citizenship, Customs & Port Security (ICP) developed the ID Card Toolkit to address the requirements of service providers (SP) to integrate, into their business applications: Identification, Authentication, Digital Signature and Non-repudiation services, around the capabilities of the Emirates ID Card.

## 1.1. Purpose

This document is a developer's reference guide to create and build the sample applications provided with the ICP ID Card Toolkit iOS and Mac SDK.

## 2. iOS and Mac Sample Application Build Instructions

### 2.1. Prerequisites

- ICP ID Card Toolkit: Objective-C and Swift SDK
- Xcode (latest version recommended)
- xmlsec (<https://www.aleksey.com/xmlsec/>)
- iPhone or iPad device (for iOS samples)
- Mac with macOS (for Mac samples)

**Note:** OpenSSL static libraries (`libcrypto.a`, `libssl.a`) are included in the iOS SDK bundle under `ext/openssl/lib/iOS/`.

**Note:** `Feitian.framework` cannot be built without the proprietary `libiRockey301_ccid_V1.31.5_release.a` supplied by Feitian under NDA. Place the file in `ext/Feitian/iOS/lib/` before building. The framework build step is skipped if the file is absent.

### 2.2. Build Instructions

#### 2.2.1. Step 1: Obtain Open-Source Libraries

OpenSSL static libraries (`libcrypto.a`, `libssl.a`) are included in the iOS SDK bundle under `ext/openssl/lib/iOS/` and do not need to be obtained separately. Get the pre-built `xmlsec` library and its header files for the target platform (iOS arm64, macOS arm64/x86\_64).

#### 2.2.2. Step 2: Open the Sample Project

Open the sample project (`.xcodproj`) in Xcode.

#### 2.2.3. Step 3: Configure Search Paths

In Xcode, go to the project settings: **Targets > Build Settings > Search Paths** and configure:

1. **Library Search Paths:** Add the directories containing the open-source libraries.
2. **Header Search Paths:** Add the directories containing the open-source library headers.
3. **Framework Search Paths:** Add the path to the Toolkit and plugin frameworks.
  - Toolkit frameworks are in the `libs` folder of the distribution. Use the Objective-C framework for Objective-C samples, or the Swift framework for Swift samples.
  - Plugin frameworks (common for both Swift and Objective-C) are in the `plugins` folder.

#### 2.2.4. Step 4: Build

Select **Product > Build** from the Xcode menu to build the sample project.

#### 2.2.5. Step 5: Connect a Device

For iOS samples, connect an iPhone or iPad to the Mac and verify it is detected in Xcode. Select the connected device from the device selector in the Xcode toolbar.

### 2.2.6. Step 6: Code Signing

1. Go to **General > Signing & Capabilities** in the project settings.
2. Add your Apple Developer account.
3. Enable **Automatically manage signing**.

### 2.2.7. Step 7: Run

Select **Product > Run** from the Xcode menu to deploy and run the sample on the connected device.

### 2.2.8. Step 8: Configure the Toolkit

On the running sample application, tap **Initialize** to configure the Toolkit. Two configuration modes are available:

#### 2.2.8.1 Option A: Bundled Configuration Files

If configuration files are bundled with the sample project:

1. Disable the URL switch.
2. Provide the following values:
  - **VG URL:** Validation Gateway service URL
  - **Config directory:** Application-specific document directory
  - **Log directory:** Application-specific document directory

**Note:** To bundle configuration files, include them in the Xcode project before building.

#### 2.2.8.2 Option B: Online Configuration URL

If configuration files are served from a URL:

1. Enable the URL switch.
2. Provide the following values:
  - **VG URL:** Validation Gateway service URL
  - **Config URL:** URL path from which configuration files can be accessed
  - **Log directory:** Application-specific document directory